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Supply network structure and firm tail risk

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ABSTRACT

This paper examines how supply network structure affects financial stability, specifically for systemic tail risks in capital markets. We construct supply networks using data from the top five clients and suppliers disclosed in the annual reports of Chinese A-share listed companies from 2009 to 2022, and measures node-level supply network structure. Our findings indicate that improved supply network structure reduces firms' systemic tail risks, primarily through information efficiency and risk diversification mechanisms. The effect is more pronounced with heightened analyst and media attention, superior corporate disclosures, closer proximity to clients or suppliers, and fewer financial constraints. This study provides insights into building resilient supply chains and mitigating financial risks.

1. Introduction

The economy is susceptible to external shocks, which can lead to the diffusion of financial risks and potentially cause systematic disruptions. Asset tail risk, in particular, has far-reaching effects and can trigger large-scale financial crises (Aloui et al., 2022). Prior research has examined various determinants of tail risk, including political uncertainty, ESG disclosure (Reber et al., 2022), and asset crowdedness (Brown et al., 2022). However, the literature has largely overlooked the role of supply chains in influencing such risks. In today's highly interconnected environment, firms are embedded in complex supply chain networks through their relationships with clients and suppliers. A key feature of these networks is their internal structure, which has been shown to significantly impact firm innovation (Bellamy et al., 2014) and investment efficiency (Shi et al., 2020). A more resilient supply chain network can also facilitate information transmission (Chuluun et al., 2014) and risk sharing (Fafchamps and Gubert, 2007; Larcker et al., 2013) among firms. This study enriches the related literature by investigating the impact of supply chain network structure on systemic tail risk.

Based on the literature in supply chain network structure, our main hypothesis is that firms with higher interconnectedness in the supply chain network incur lower systematic tail risks. There are two primary reasons. First, firms with a robust network structure tend to have a larger number of clients and suppliers, which facilitates the flow of information. This enhanced information transmission enables firms to obtain valuable insights more efficiently (Chuluun et al., 2014) and accelerates price discovery. During adverse events, firms with higher information efficiency are better able to incorporate private information (Jin and Myers, 2006), making them less vulnerable to investor sell-offs. Second, a well-connected supply chain network involves more partners, which reduces contracting

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costs and simplifies the establishment of agreements among firms (Larcker et al., 2013). When facing credit constraints, these firms are more likely to receive support from their network partners, thereby lowering their default risk. Overall, a more interconnected and resilient supply chain network structure can help firms mitigate systemic tail risk during severe market downturns.

We utilize data from Chinese A-share listed companies on the Shanghai and Shenzhen Stock Exchanges from 2009 to 2022. Using disclosed information on the top five customers and suppliers from annual reports, we construct supply chain network structures to examine their impact on financial stability, with a particular focus on systemic tail risk. Our empirical evidence suggests that stronger supply chain network structures are associated with lower firm-level systemic tail risk. We further demonstrate that this effect operates primarily through two mechanisms: improved information efficiency and enhanced risk diversification. Heterogeneity analyses reveal that the mitigating effect of supply chain network structure on systemic tail risk is more pronounced in firms with greater analyst and media coverage, higher-quality accounting disclosures, closer geographic proximity to customers or suppliers, and lower financial constraints. Additionally, we perform a series of robustness checks, including the PSM-DID approach, the Heckman two-stage model, and alternative variable constructions. The main results remain consistent and robust across these tests.

Our study contributes to the literature on supply chain network structure. While prior research has extensively examined the conceptualization, characteristics, and determinants of supply chain networks (Ponis and Koronis, 2012; Chopra and Sodhi, 2014; Hosseini et al., 2019), relatively little attention has been paid to the implications of structural features within these networks. Bellamy et al. (2014) show that supply chain network structure influences corporate innovation, and Shi et al. (2020) find that firms located at the center of supply networks are better able to mitigate underinvestment. Wu and Birge (2014) examine how a firm's centrality within the supply chain affects stock returns, particularly across industries. However, the impact of supply chain network structure on firm-level risk, especially systemic tail risk, and the mechanisms through which it operates remain underexplored. By focusing on Chinese listed companies, our study fills this gap by providing novel empirical evidence on how supply chain network structure influences corporate tail risk. In doing so, we contribute to a deeper understanding of the intersection between supply chain dynamics and financial market stability.

This paper also offers a novel perspective on the determinants of firm-level tail risk. Prior studies have identified various drivers of tail risk, including political uncertainty, ESG disclosure (Reber et al., 2022), and asset crowdedness (Brown et al., 2022). In contrast, we focus on the role of supply chain network structure, recognizing that firms are embedded in interconnected supply networks and are closely linked to their peers. Our findings show that a more robust supply chain network structure significantly improves information efficiency and enhances risk diversification, thereby reducing systemic tail risk.

The structure of the paper is organized as follows: Part 2 presents the literature review and hypothesis formulation. Part 3 describes the data, variables, and research design. Part 4 provides the empirical results. Part 5 investigates the mechanisms and conducts heterogeneity tests. Part 6 concludes.

2. Literature review and hypothesis development

2.1. Supply chain network structure

In recent years, a growing body of literature has examined the roles of social networks in information dissemination, resource acquisition, risk sharing, behavior shaping, and the development of group trust (Larcker et al., 2013; Bianchi et al., 2022; Wu et al., 2024). Scholars have constructed social networks involving directors, CEOs, analysts, auditors, and investors based on various types of cross-firm relationships, providing empirical evidence of their influence on corporate operations (Hochberg et al., 2007, 2010; Larcker et al., 2013; El-Khatib et al., 2015; Ahern, 2017; Bianchi, 2018; Rossi et al., 2018).

Within the supply chain network structure, suppliers, manufacturers, service providers, alliance partners, and customers are connected through activities related to the procurement, production, and delivery of goods and services (Kouvelis et al., 2006; Lamming et al., 2000). Prior research suggests that network participants benefit from both direct and indirect connections (Granovetter, 1973; Burt, 1995) as well as their positions within the network (Adler and Kwon, 2002; Borgatti and Foster, 2003). Firms with a stronger supply chain network structure possess greater social resources, enabling them to adopt risk-mitigating strategies such as enhancing supply chain collaboration (Pettit et al., 2013), making better supplier selections (Hosseini et al., 2019), eliminating redundancies (Ponis and Koronis, 2012), and actively fostering social capital and relationships (Johnson et al., 2013). Empirical studies have examined how the structure of supply chain networks affects firm innovation (Bellamy et al., 2014), opportunism (Dong et al., 2015), productive efficiency (Kao et al., 2017), profitability (Kim, 2017), and investment efficiency (Shi et al., 2020). However, the literature has largely overlooked the impact of supply chain network structure on systemic risk. Our paper addresses this gap by providing empirical evidence on how supply chain network structure influences systemic tail risk.

2.2. Systemic tail risk

Systemic tail risk refers to the risk of substantial asset losses during periods of extreme market downturns. Compared to general financial risks, systemic tail risk has broader economic implications and can trigger large-scale financial crises. Scholars have increasingly recognized systemic tail risk as an important factor in risk pricing (Kelly and Jiang, 2014; Van Oordt and Zhou, 2016; Chabi-Yo et al., 2018). Kelly and Jiang (2014) hypothesize that the left tail of the asset return distribution follows a dynamic power law process and find that stocks with higher systemic tail risk in the U.S. market tend to generate higher future returns. Van Oordt and Zhou (2016) develop a tail beta index to measure the sensitivity of asset prices to extreme market risks, demonstrating significant risk premia associated with tail beta. Chabi-Yo et al. (2018) employ a mixture of dynamic copula functions to assess individual stock sensitivity to

market crashes, revealing that lower tail dependence coefficients have strong explanatory power for cross-sectional stock returns.

Other studies have explored the drivers of systemic tail risk, highlighting the impacts of political uncertainty, ESG disclosure (Reber et al., 2022), and asset crowdedness (Brown et al., 2022). For example, Kelly et al. (2016) examine national elections and global summits and find that political uncertainty is priced in the options market, with option protection against tail risk becoming more expensive before political events. Reber et al. (2022) analyze IPOs in the U.S. and demonstrate that voluntary ESG disclosure reduces both idiosyncratic volatility and downside tail risk. Furthermore, Brown et al. (2022) show that hedge fund exposures related to crowdedness significantly explain downside tail risk. Additionally, under the impact of Black Swan events, tail risk spillovers among RMB exchange rates exhibit significant dynamic and time-varying characteristics (Huang and Zhang, 2024). Building on the approach of Kelly and Jiang (2014), this study estimates systemic tail risk and addresses a gap in the literature by examining the determinants of systemic risk from the perspective of supply chain network structure.

2.3. The impact of supply chain network structure on systemic tail risks

This study posits that the supply chain network structure helps mitigate systemic tail risk through two primary mechanisms. First, networks serve as channels for aggregating and disseminating various types of information (Freeman, 2002). Firms with a robust network structure maintain numerous client and supplier relationships that facilitate information flow, allowing them to access valuable information more easily (Chuluun et al., 2014). Consequently, information is incorporated into stock prices more rapidly. During extreme market events, firms with higher information efficiency better reflect private information (Jin and Myers, 2006), making them less susceptible to large investor sell-offs and thus reducing the likelihood of systemic tail risk.

Second, from the perspective of risk diversification, a strong supply chain network structure includes many potential stakeholders, enabling more resource sharing through financial transactions with other companies. Studies such as Larcker et al. (2013) indicate that board networks provide risk insurance for firms, facilitating contract establishment and reducing contracting costs among firms. When facing liquidity constraints or inadequate credit allocation, firms with a stronger supply chain network structure are more likely to receive assistance from other network members, thereby dispersing individual default risks, thereby lowering systemic tail risks as the market experiences severe downturns. Hence, we propose the following hypothesis H_1:

H_1. : Within supply chain networks, firms with stronger supply chain network structure exhibit lower systemic tail risks.

3. Data and variables

3.1. Data source and sample selection

To conduct empirical analyses, we begin with a sample of all A-share Chinese listed firms from 2009 to 2022. Supply chain network data comes from the China Research Data Services Platform (CNRDS). Monthly stock trading data and company financial statements are obtained from the China Stock Market & Accounting Research (CSMAR) database. We exclude financial and ST (Special Treatment) firms and remove firms with missing data in our sample. Our final sample consists of 9414 firm-year observations. To mitigate the influence of outliers, all continuous variables are winsorized at the 1st and 99th percentiles.

3.2. Variables definitions

3.2.1. Dependent variable: systemic tail risk

This study uses systemic tail risk as a proxy for financial stability to investigate the impact of supply chain network structure. Following the methodologies of Kelly and Jiang (2014), systemic tail risk is measured as follows: First, for the return sequence of stock i, which is denoted as $\{r_{i,t}\}_{t=1}^{T}$, given the threshold u_t, and based on the information set F_t, we assume its left-tail distribution follows a dynamic power-law process. Therefore, the conditional probability of left-tail returns satisfies:

$$P\left(r_{i,t+1} < r \middle| r_{i,t+1} < u_t and F_t\right) = \left(\frac{r}{u_t}\right)^{-a_i/\lambda_t}, r < u_t < 0\# \tag{1}$$

Where $r_{i,t+1}$ represents the return of stock i on the $t+1$-th trading day, F_t is the information set at time t, and $r < u_t < 0$. The parameters $a_i$andλ_t collectively determine the shape of the power-law tail distribution. A larger value of λ_t indicates a greater tail risk, resulting in a thicker left-tail distribution of returns and thus a higher probability of tail risk occurrence.

Furthermore, the overall tail risk of the stock market at time t is measured using the power-law method proposed by Hill(1975):

$$\lambda_t^{Hill} = \frac{1}{K_t}\sum_{k=1}^{K_t} \ln\frac{r_{k,t}}{u_t} \tag{2}$$

Where u_t represents the threshold level set in month t, $r_{k,t}$ denotes the k-th daily return of a stock below the threshold level u_t in the month t, and K_t represents the total number of daily returns below the threshold level u_t for all stocks in month t. Following Kelly and Jiang (2014), we set the threshold as the fifth percentile of the cross-section distribution of the stocks' returns each period.

Finally, the systemic tail risk of stock i in month $_t$ is estimated using a rolling window over the past 36 months:

$$r_{i,t} = \alpha_i + Tr_i\lambda_t^{Hill} + \varepsilon_{i,t} \tag{3}$$

Table 1
Topological indicators of the supply network.

Year	Node	Edge count	Average outdegree	Average indegree	Density	Number of connected components
2009	5149	4692	0.911	0.911	0.0002	538
2010	6689	6211	0.929	0.929	0.0001	625
2011	8120	7673	0.945	0.945	0.0001	668
2012	9589	9361	0.976	0.976	0.0001	600
2013	8790	8504	0.967	0.967	0.0001	581
2014	6000	5736	0.956	0.956	0.0002	430
2015	6554	6322	0.965	0.965	0.0001	396
2016	5810	5654	0.973	0.973	0.0002	336
2017	5619	5460	0.972	0.972	0.0002	325
2018	4865	4743	0.975	0.975	0.0002	289
2019	4552	4421	0.971	0.971	0.0002	299
2020	4301	4155	0.966	0.966	0.0002	301
2021	5562	5304	0.954	0.954	0.0002	481

This table provides the statistics for topological indicators of the supply network from 2009 to 2021 for our sample, which include the node, edge count, the average outdegree, the average indegree, density and number of connected components.

Where $r_{i,t}$ represents the monthly return of stock i in month t, α_i is the intercept term, λ_t^{Hill} is the estimated overall market tail risk from Eq. (2), and Tr_i denotes the systemic tail risk for stock i. A larger Tr_i indicates higher systemic tail risk for the stock.

3.2.2. Independent variable: supply chain network structure

Common indicators used to characterize a firm's position within networks include various measures of network centrality, such as degree centrality, closeness centrality, betweenness centrality, and eigenvector centrality. Among these, degree centrality is a primary measure of the importance of a node within a network, quantifying the number of connections a node has with other nodes. Firms with higher degree centrality are closer to the core of the network. Given large heterogeneity in connections between nodes, prior papers also measure weighted degree centrality, where weights are calculated based on the strength of connections between nodes. Drawing on approaches by Kim et al. (2011) and Allen et al. (2019), our study computes the weighted degree centrality in the supply chain network to gauge supply chain network structure.

Specifically, we compile two panel datasets from 2009 to 2022 for both the top five customer firms and top five supplier firms of listed companies, and construct directed weighted networks of customer-supplier relationships on an annual basis. Assuming the supply chain network is a directed weighted graph containing m firms, represented by elements $0 \leq w_{i,j} \leq 1$ of the $m \times m$ matrix, where $w_{i,j} > 0$ denotes the total sales amount of products provided by firm i to enterprise j. We normalize the sales amount and use the NetworkX software package in Python to compute the weighted degree centrality. We then take the natural logarithm of these centrality values to measure supply chain network structure (Sns) , where $w_{p,q,t}$ represents the strength of connections between firms in the following equation:

$$Sns_{i,t} = \ln\left(1 + \sum\nolimits_{p \neq q} w_{p,q,t}\right), p \in iorq \in i \tag{4}$$

To further obtain the dynamic evolution of the supply chain network, we annually calculate network topology metrics such as node and edge counts, average in-degree and out-degree, number of connected components, and network density.Table 1[1] presents various supply chain network topology indicators from 2009 to 2021. The trends observed in China's supply chain network are: (1) the node and edge counts display an initial increase followed by a decline, indicating a trend towards a stable state as the supply chain network expands and becomes more intricate; (2) the average in-degree and out-degree exhibit an overall upward trajectory but remain below 1, suggesting that the majority of firms typically engage with either a single supplier or customer; (3) while network density experiences a modest increase, the number of connected components continues to diminish, signifying a gradual shift towards an asymmetric distribution of firm network positions. In summary, an increasing number of firms are integrating into established industry chains through supply chain relationships, leading to more direct and indirect interconnections among firms, which echoes with prior research.

3.2.3. Control variables

Following the methodology of existing literature, we add firm-level characteristics into the regression model. These basic characteristics include the natural logarithm of total market capitalization (*Size*), book-to-market ratio (*Mb*), and return on equity (*Roe*). We also incorporate stock-specific vulnerability features, including stock price (*Price*), previous-period stock returns (*Pastret*), and illiquidity ratio (*Illiq*). Moreover, the model controls for other factors affecting systemic tail risk, including stock-specific volatility (*Ivol*). To mitigate the potential impact of cross-sectional heterogeneity and varying time trends on the regression results, we account for firm- and year-fixed effects. Detailed definitions for variables are provided in Table 2.

[1] Because the dependent variable is lagged by one period, Table 1 displays supply network topology indicators up to 2021.

Table 2
Variable definitions.

Variables	Definitions
Tr	Systemic tail risk, a systemic tail risk indicator for the month corresponding to the year-end. Please refer to the Section 3.2.1 for detailed elaboration.
Sns	Supply network structure, the natural logarithm of the weighted degree centrality of nodes in a supply network
Size	The natural logarithm of the total market capitalization of stocks.
Mb	Market capitalization divided by shareholder equity.
Roe	Net income divided by net assets.
Price	The natural logarithm of the year-end stock price.
Pastret	The stock return of the previous fiscal year.
Illiq	Amihud(2002) stock illiquidity measure.
Ivol	The adjusted trading-day-based standard deviation of stock residual returns calculated using the FF3 factor model.

This table reports the definitions of the main variables.

Table 3
Descriptive Statistics.

Variables	Observations	Mean	Standard deviation	Minimum	Median	Maximum
Tr	9414	0.597	0.394	−0.423	0.542	1.665
Sns	9414	0.022	0.055	0.00001	0.004	0.377
Size	9414	22.573	1.013	20.843	22.417	25.792
Mb	9414	3.495	3.447	0.534	2.496	24.021
Roe	9414	0.050	0.156	−0.946	0.063	0.320
Price	9414	2.300	0.668	0.928	2.247	4.065
Pastret	9414	0.145	0.645	−0.734	−0.019	2.688
Illiq	9414	0.050	0.049	0.002	0.035	0.282
Ivol	9414	0.327	0.097	0.132	0.320	0.594

This table reports the descriptive statistics of main variables for our sample between 2009 and 2021. Specific variable definitions are in Table 2.

Table 4
The impact of supply network structure on systemic tail risk.

	(1) *Tr*	(2) *Tr*	(3) *Tr*	(4) *Tr*	(5) *Tr*
Sns	−0.502***	−0.456***	−0.299***	−0.354***	−0.219*
	(−4.89)	(−4.49)	(−2.78)	(−3.54)	(−1.80)
Size		−0.013	−0.019	−0.013	−0.023
		(−0.82)	(−1.20)	(−0.80)	(−1.15)
Mb		−0.002	−0.004	−0.003	−0.008**
		(−1.00)	(−1.46)	(−1.17)	(−2.27)
Roe		−0.043	−0.061*	−0.031	−0.015
		(−1.26)	(−1.83)	(−0.89)	(−0.36)
Price		0.027	0.026	0.017	0.029
		(1.62)	(1.54)	(1.04)	(1.38)
Pastret		−0.065***	−0.074***	−0.072***	−0.080***
		(−6.91)	(−8.01)	(−7.62)	(−7.26)
Illiq		0.243**	0.146	0.200*	0.178
		(2.09)	(1.26)	(1.67)	(1.35)
Ivol		0.057	0.053	0.038	0.040
		(1.10)	(1.02)	(0.73)	(0.64)
Cons	0.608***	0.821**	0.964***	0.847**	1.070**
	(272.22)	(2.50)	(2.94)	(2.56)	(2.55)
Year FE	Yes	Yes	No	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes
Year × Ind FE	No	No	Yes	No	No
Year × Pro FE	No	No	No	Yes	No
Year × Ind × Pro FE	No	No	No	No	Yes
R^2	0.594	0.599	0.634	0.630	0.733
adj. R^2	0.503	0.509	0.538	0.522	0.561
N	9414	9414	9414	9414	9414

This table examines the impact of supply network structure on systemic tail risk. The dependent variable is *Tr*, which is a systemic tail risk indicator for the month at the year-end. The independent variable is *Sns*, which is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

3.3. Model specification

To examine the impact of supply chain network structure on systemic tail risk, we employ the following baseline OLS equation:

$$Tr_{i,t+1} = \alpha + \beta Sns_{i,t} + \gamma Controls_{i,t} + \theta_t + \psi_i + \varepsilon_{i,t+1} \tag{5}$$

where i denotes the firm and t represents the year. $Tr_{i,t+1}$ represents the systemic tail risk of firm i in period $t+1$. $Sns_{i,t}$ denotes the supply chain network structure for firm i in period t, calculated using weighted degree centrality. $Controls_{i,t}$ represents a series of control variables. θ_t denotes year fixed effects, ψ denotes firm fixed effects, and $\varepsilon_{i,t+1}$ represents the error term. To address potential issues of serial correlation, we cluster the standard errors at the firm level.

4. Empirical results

4.1. Descriptive statistics

Table 3 presents descriptive statistics for the key variables in this study. The systemic tail risk indicator (*Tr*) exhibits a mean of 0.597, with a range from −0.423–1.665, and a standard deviation of 0.394, indicating substantial cross-sectional variation in systemic tail risk levels. The supply chain network structure indicator (*Sns*) shows a mean of 0.022 and a standard deviation of 0.055, with a notable difference between the minimum and maximum values, highlighting significant variability in network structure across companies. Moreover, the descriptive statistics for the control variables are consistent with findings from existing studies.

4.2. Baseline regression results

Table 4 examines the impact of supply chain network structure on systemic tail risk. Column (2) expands upon Column (1) by including all control variables. To further validate the baseline regression results against potential omitted variable bias, we include high-dimensional fixed effects such as year-by-industry interactions, year-by-province interactions, and triple interactions of year, province, and industry. The coefficient on supply chain network structure remains significantly negative at the 1 % or 10 % level, indicating that stronger supply chain network structure is associated with lower systemic tail risk in the future, supporting hypothesis H_1. Furthermore, Column (2) reveals that a one standard deviation increase in supply chain network centrality makes the systemic tail risk decrease by 4.2 %,[2] which indicates great economic significance.

4.3. Robustness checks

4.3.1. Endogeneity issues

To mitigate endogeneity issues, we first employ the Propensity Score Matching-Double Difference (PSM-DID) model, leveraging the "Chain Leader" policy shock.[3] The "Chain Leader", proposed by Hunan Province in 2017 and implemented province-wide in Zhejiang Province in 2019, had been adopted by 29 provinces (autonomous regions, municipalities directly under the central government) in mainland China by December 2021. This policy aims to enhance industrial competitiveness and improve supply chain network structure through strategic initiatives.

The implementation of the “Chain Leader” policy is typically centrally and locally coordinated by the government. Its timing and scope are related to the supply chain network structure but are not endogenously linked to the risk characteristics of individual enterprises. Thus, it can be considered an effective exogenous policy shock. Specifically, the Chain Leader policy is usually implemented in a top-down manner by local or central governments, and its policy motivation is unrelated to the micro characteristics of individual firms. Instead, it is more focused on macro-objectives such as regional industrial planning and supply chain security. Additionally, the timing of policy implementation is exogenous to firms, as they cannot predict or adjust their supply chain strategies in advance. The policy is typically implemented at the industry or regional level, rather than targeting specific companies, thus avoiding endogeneity issues that would arise if firms actively chose to enter the treatment group.

[2] $(-0.456 * 0.055)/0.597 * 100\% = -4.2\%$

[3] The implementation of the “Chain Leader” policy can be roughly divided into three stages. The first stage is the local autonomous exploration period. In 2017, Changsha City in Hunan Province first implemented the Chain Leader System. The key feature of this stage was that each region, based on its own endowments, independently explored a “Chain Leader” model suitable for local industrial development. The main goal of the policy at this stage was to enhance the competitiveness of local industrial chains and supply chains, although the policy’s reach in terms of regions and industries was relatively limited. The second stage was the gradual promotion period. In 2019, Zhejiang Province became the first province to implement the Chain Leader System across the entire province. The key feature of this stage was the gradual expansion of the policy’s implementation scope, with the main goal shifting towards improving the decoupling and disruption of industrial chains caused by external factors. The third stage was the comprehensive deepening period. In September 2020, the State Council issued the “Notice on Issuing the Overall Plan for the Beijing, Hunan, and Anhui Free Trade Zones and the Expanded Area Plan for the Zhejiang Free Trade Zone” (State Council Document [2020] No. 10). By this point, the Chain Leader System shifted from local active exploration to full nationwide deployment. The key feature of this stage was the nationwide expansion of the policy, aiming to improve the resilience and security of industrial and supply chains, focusing on building, supplementing, and extending key industrial chains.

Table 5
Propensity score matching balance test.

Covariates	Unmatched(U)/Matched (M)	Treatment group	Control group	Percentage deviation	Percentage reduction in deviation	T-value	P-value
Size	U	22.724	22.626	10.00		7.02	0.000
	M	22.723	22.716	0.70	93.10	0.39	0.697
Mb	U	3.498	3.476	0.70		0.47	0.640
	M	3.504	3.527	−0.70	−5.00	−0.42	0.672
Roe	U	0.031	0.043	−6.30		−4.52	0.000
	M	0.032	0.024	4.50	28.30	2.35	0.019
Price	U	2.440	2.369	9.10		6.55	0.000
	M	2.441	2.428	1.60	81.90	0.92	0.359
Pastret	U	0.174	0.076	19.90		13.50	0.000
	M	0.171	0.180	−1.70	91.40	−0.89	0.373
Illiq	U	0.036	0.049	−30.40		−19.00	0.000
	M	0.036	0.037	−2.40	92.10	−1.64	0.100
Ivol	U	0.387	0.325	58.00		41.47	0.000
	M	0.385	0.386	−0.70	98.70	−0.39	0.698

This table presents the balance test for propensity score matching, showing that after matching, the standardized mean differences for most paired variables are below 10 %. Additionally, there are almost no significant differences between the experimental and control groups regarding control variables, basically meeting the balance test requirements.

Table 6
Heckman two-stage regression.

	(1) *Tr*	(2) *Disclosure*	(3) *Tr*
Ev_dis		2.407***	
		(12.56)	
Sns			−0.598***
			(−5.76)
Imr			0.026
			(0.50)
Post	−0.025**		
	(−2.00)		
Size	−0.005	−0.083***	−0.021
	(−0.28)	(−4.21)	(−1.20)
Mb	0.002	0.019***	−0.002
	(0.63)	(3.54)	(−0.63)
Roe	−0.005	0.053	−0.037
	(−0.21)	(0.83)	(−1.06)
Price	0.006	−0.169***	0.015
	(0.32)	(−5.95)	(0.82)
Pastret	0.004	0.020	−0.069***
	(0.43)	(1.23)	(−6.50)
Illiq	0.334	−0.043	0.069
	(1.62)	(−0.34)	(1.05)
Ivol	0.198***	0.375***	0.057
	(4.43)	(2.87)	(0.99)
Cons	0.874**	0.263	1.394***
	(2.18)	(0.61)	(3.99)
Year FE	Yes	Yes	Yes
Firm FE	Yes		Yes
R^2	0.311		0.384
Pseudo. R^2/*adj.* R^2	0.310	0.183	0.382
N	12360	37920	8581

This table reports the results of Heckman two-stage regression. In the first stage, the dependent variable is "Disclosure of Supplier or Customer Information (*Disclosure*)" (a dummy variable where disclosure is 1 and non-disclosure is 0). The mean disclosure rate of companies within the same industry (Ev_dis) is included as an exclusion restriction variable. *Tr* is a systemic tail risk indicator for the month at the year-end. *Sns* is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

Using data from 2012 to 2022, this study constructs the PSM-DID model, covering five years before and five years after the policy implementation. If a firm operates in a province implementing the "Chain Leader" policy in a given year, the variable Post is coded as 1; otherwise, it is 0. This variable represents the interaction between the experiment group and the treatment period. We employ 1:1 nearest neighbor logit matching with a caliper of 0.001, using the same set of control variables as the baseline model. This process

Table 7
Portfolio tests of systemic tail risk indicator.

Portfolio	(1) *RAWRET*	(2) *CAPM*	(3) *FF3*	(4) *CHN3*	(5) *CHN4*
LOW	0.530	−0.209	−0.397**	−0.011	−0.049
	(0.79)	(−0.62)	(−2.44)	(−0.06)	(−0.28)
2	0.862	0.123	−0.112	0.119	0.041
	(1.29)	(0.38)	(−0.89)	(0.84)	(0.31)
3	1.071	0.330	0.082	0.275**	0.201*
	(1.60)	(1.00)	(0.77)	(2.25)	(1.80)
4	1.128	0.375	0.116	0.240**	0.177*
	(1.65)	(1.13)	(1.10)	(2.30)	(1.78)
5	1.177*	0.425	0.159	0.317***	0.277***
	(1.70)	(1.21)	(1.43)	(3.14)	(2.77)
6	1.275*	0.507	0.228**	0.436***	0.427***
	(1.78)	(1.38)	(2.17)	(4.46)	(4.24)
7	1.311*	0.539	0.266**	0.464***	0.457***
	(1.80)	(1.41)	(2.25)	(4.68)	(4.37)
8	1.140	0.353	0.067	0.302***	0.326**
	(1.50)	(0.85)	(0.49)	(2.63)	(2.61)
9	1.243	0.452	0.154	0.451***	0.483***
	(1.60)	(1.03)	(1.03)	(3.36)	(3.39)
HIGH	1.172	0.371	0.090	0.616***	0.666***
	(1.42)	(0.73)	(0.48)	(3.26)	(3.38)
H-L	0.641*	0.581*	0.487*	0.627**	0.715**
	(1.91)	(1.75)	(1.71)	(2.03)	(2.28)

This table reports the results of portfolio tests to further investigate the relationship between systemic tail risk and cross-sectional stock returns. Specifically, at the beginning of each month, all stocks are divided into 10 groups based on the size of their systemic tail risk indicators. Using equal-weighting, the highest-ranked group is longed while the lowest-ranked group is shorted to construct a stock hedge portfolio, held for the following month.

results in 6180 observations each for the experiment and control groups, totaling 12,360 observations.

Table 5 presents the balance test for propensity score matching, showing that after matching, the standardized mean differences for most paired variables are below 10 %. Additionally, there are almost no significant differences between the experiment and control groups in terms of control variable characteristics, meeting the balance test requirements. In Table 6, Column (1) displays the results of the PSM-DID regression. The coefficient on Post is significantly negative at the 5 % level, indicating that improving supply chain network structure contributes to reducing future systemic tail risk for firms, thus further supporting our hypothesis.

However, in annual reports, due to the non-mandatory nature of corporate disclosure regarding customer and supplier information, our main conclusions may be subject to sample self-selection bias. To address this issue, we employ the Heckman two-stage regression. In the first stage, we use "Disclosure of Supplier or Customer Information (Disclosure)" (a dummy variable where disclosure is 1 and non-disclosure is 0) as the dependent variable. The mean disclosure rate of companies within the same industry (Ev_dis) is included as an exclusion restriction variable. Control variables are the same with Model (5). The regression estimates the inverse Mills ratio (IMR), which is subsequently incorporated into the second-stage regression. In Table 6, Column (3) presents the results of the Heckman two-stage regression. After controlling for sample self-selection bias, the coefficient on supply chain network structure remains significantly negative at the 1 % level, which indicates the robustness of the main findings.

4.3.2. Testing systemic tail risk indicators

The validity of our main findings relies on the effectiveness of the time-varying systemic tail risk measure, proposed by Kelly and Jiang (2014), to accurately capture the dynamic changes in systemic tail risk in the Chinese A-share market. To assess this, we employ portfolio tests to investigate the predictability of systemic tail risk on cross-sectional stock returns.[4] Specifically, at the beginning of each month, all stocks are divided into 10 groups based on the size of their systemic tail risk indicators. Using equal-weighting, we construct a stock hedge portfolio by longing the highest-ranked group and shorting the lowest-ranked group, holding this portfolio for the following month. Table 7 demonstrates that portfolios with higher systemic tail risk expectations generally outperform those with lower systemic tail risk expectations, even accounting for risk factors. The stock hedge portfolios yield significant excess returns. These results corroborate the validity of using the systemic tail risk method for the Chinese market. Notably, traditional three-factor model calculations may capture spurious excess returns driven by market volatility or style rotations, whereas incorporating localized size and value factors tailored to the Chinese market could more objectively identify the true drivers of excess returns in this context, thereby refining asset allocation decisions. Therefore, following Liu et al. (2019), we integrate China-specific size and value factors into the Fama-French framework. This approach mitigates biases from market noise and style shifts, as shown in Columns (4)-(5) of Table 7, where our conclusion still holds.

[4] The systematic tail risk indicators in this study begin from the end of 2010, hence the portfolio testing period spans from January 2011 to December 2022.

Table 8
Alternative systematic tail risk measures.

	(1) $u_t = 3\%$	(2) $u_t = 10\%$	(3) *Window* = 48	(4) *Window* = 60
	Tr	*Tr*	*Tr*	*Tr*
Sns	−0.589***	−0.189**	−0.463***	−0.147**
	(−5.11)	(−1.98)	(−5.09)	(−2.11)
Size	0.012	−0.011	−0.012	−0.028**
	(0.76)	(−0.73)	(−0.89)	(−2.32)
Mb	−0.001	−0.008***	−0.003*	−0.002
	(−0.49)	(−3.57)	(−1.82)	(−1.28)
Roe	−0.043	−0.109***	−0.027	−0.042*
	(−1.34)	(−3.09)	(−1.01)	(−1.84)
Price	0.019	0.011	0.048***	0.056***
	(1.14)	(0.67)	(3.24)	(4.16)
Pastret	−0.121***	−0.004	−0.017***	−0.002
	(−12.86)	(−0.49)	(−2.62)	(−0.30)
Illiq	0.155	0.067	0.317***	0.176*
	(1.26)	(0.62)	(3.26)	(1.89)
Ivol	0.044	0.373***	0.028	−0.065*
	(0.81)	(6.60)	(0.66)	(−1.77)
Cons	0.201	0.826**	0.752***	1.100***
	(0.61)	(2.52)	(2.59)	(4.27)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
R^2	0.617	0.624	0.623	0.656
adj. R^2	0.530	0.540	0.538	0.579
N	9414	9414	9414	9414

This table reports the results of the regressions after replacing the dependent variable with different indicators. *Sns* is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

4.3.3. Alternative variable measurement

To ensure the robustness of our results, we first change the measurement of systemic tail risk. The selection of thresholds is crucial for estimating systemic tail risk. A threshold that is too high can result in a large number of non-extreme observations, while one that is too low may lead to a small number of observations. Therefore, we recalculate systemic tail risk using threshold levels of third and tenth percentile. Additionally, the choice of rolling regression windows can significantly affect the estimation of systemic tail risk due to variations in the number of valid observations. Hence, we reassess systemic tail risk using rolling windows of 48 months and 60 months. Table 8 presents the corresponding regressions results. The findings consistently align with the main hypothesis.

Next, we change the measurement of supply chain network structure. In the baseline regression, supply chain network structure is measured using the natural logarithm of weighted degree centrality. For robustness checks, we employ alternative centrality measures, including the natural logarithm of node degree centrality (*Lndegree*, the natural logarithm of the number of directly connected nodes), standardized node degree centrality (*Degree_centrality*, the number of connected nodes divided by total nodes minus one), betweenness centrality (*Betweenness*), and closeness centrality (*Closeness*), all normalized (Z-scored) for comparability. Table 9 presents the corresponding regression results. The findings are consistent with those of the baseline regression, indicating that the main conclusions remain robust with different measurements of the explanatory variable.

4.3.4. Other robustness checks

Our findings remain robust by: (1) Changing the sample interval. On September 19, 2012,[5] the China Securities Regulatory Commission issued an announcement that encourages listed companies to disclose the names and procurement amounts of their top five customers and suppliers, starting from January 1, 2013. We thus narrow the sample period to the years between 2013 and 2022. (2) Non-linear relationship examination. Given the potential non-linear relationship between supply chain network structure and systemic tail risk, we adopt the approach of Bellamy et al. (2014) by including the square term of supply chain network structure in the baseline model regression. The results indicate that the coefficient of the network structure squared term is not statistically significant. (3) Other forms of social networks. It is important to consider the impact of other forms of social networks, such as board member networks, on a company's risk. Board members' personal social networks serve as informal channels for the transfer of information and resources, thus influencing corporate risk decisions. We further include board network centrality as control variable in the model. The

[5] China Securities Regulatory Commission Announcement [2012] No. 22, 'Guidelines No. 2 on Content and Format of Information Disclosure for Companies Issuing Securities - Content and Format of Annual Reports.'

Table 9
Alternative measures for supply network structure.

	(1) *Tr*	(2) *Tr*	(3) *Tr*	(4) *Tr*
Lndegree	−0.021***			
	(−3.43)			
Degree_centrality		−0.024***		
		(−3.47)		
Betweenness			−0.023***	
			(−3.01)	
Closeness				−0.025***
				(−3.94)
Size	−0.017	−0.017	−0.017	−0.016
	(−1.07)	(−1.07)	(−1.08)	(−1.03)
Mb	−0.002	−0.002	−0.002	−0.002
	(−1.00)	(−0.98)	(−1.02)	(−0.98)
Roe	−0.045	−0.045	−0.044	−0.044
	(−1.33)	(−1.32)	(−1.28)	(−1.30)
Price	0.030*	0.029*	0.030*	0.029*
	(1.83)	(1.74)	(1.79)	(1.72)
Pastret	−0.067***	−0.067***	−0.066***	−0.067***
	(−7.08)	(−7.09)	(−6.99)	(−7.02)
Illiq	0.248**	0.239**	0.245**	0.238**
	(2.12)	(2.05)	(2.10)	(2.03)
Ivol	0.066	0.070	0.069	0.068
	(1.28)	(1.35)	(1.34)	(1.31)
Cons	0.890***	0.893***	0.887***	0.877***
	(2.70)	(2.70)	(2.71)	(2.68)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
R^2	0.599	0.599	0.598	0.599
adj. R^2	0.508	0.508	0.508	0.508
N	9414	9414	9414	9414

This table presents the results of regressions after replacing the explanatory variable with different measures, including the natural logarithm of node degree centrality (Lndegree, the natural logarithm of the number of directly connected nodes), standardized node degree centrality (Degree_centrality, the number of connected nodes divided by total nodes minus one), betweenness centrality (Betweenness), and closeness centrality (Closeness). Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

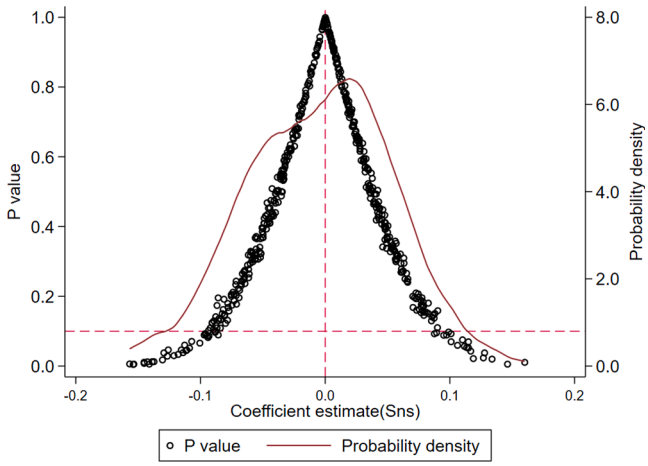

Fig. 1. Placebo Test.

Table 10
Additional robustness tests.

	(1) Shorten the sample time interval	(2) Testing for non-linear relationships	(3) Controlling the impact of the board network
	Tr	*Tr*	*Tr*
Sns	−0.190*	−0.307	−0.440***
	(−1.78)	(−1.18)	(−4.20)
Sns^2		−0.450	
		(−0.61)	
Board_centrality			−0.008
			(−0.48)
Size	−0.033	−0.013	−0.014
	(−1.55)	(−0.86)	(−0.88)
Mb	0.000	−0.002	−0.003
	(0.06)	(−0.99)	(−1.28)
Roe	−0.050	−0.043	−0.042
	(−1.29)	(−1.26)	(−1.22)
Price	0.043**	0.027	0.032*
	(2.01)	(1.64)	(1.87)
Pastret	−0.006	−0.065***	−0.063***
	(−0.60)	(−6.91)	(−6.82)
Illiq	0.198	0.244**	0.256**
	(1.43)	(2.09)	(2.17)
Ivol	0.049	0.058	0.053
	(0.88)	(1.12)	(1.00)
Cons	1.117**	0.833**	0.836**
	(2.48)	(2.52)	(2.51)
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
R^2	0.600	0.599	0.603
adj. R^2	0.488	0.509	0.512
N	5732	9414	9006

This table reports the regression results by changing the sample period interval, conducting non-linear relationship test and controlling the impact of the board network. The dependent variable is *Tr*, which is a systemic tail risk indicator for the month at the year-end. The independent variable is Sns, which is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

results show that our conclusions still hold. (4) Placebo tests. The placebo test can verify the robustness of the results from a counterfactual perspective, ruling out structural confounding factors that may not be captured by high-dimensional fixed effects.[6] Specifically, we perform 500 random matching placebo tests and find that the estimated coefficient of *Sns* in the baseline regression (-0.456) lies on the left side of the placebo test coefficient distribution, which is a very low-probability event. This means that in the random trials, the supply chain network structure does not show a significant effect on systemic tail risk, further validating the robustness of our main conclusions (see Fig. 1 below).

5. Underlying mechanisms and heterogeneity analysis

In the previous analyses, we demonstrate that enhancing supply chain network structure helps reduce future systemic tail risks for firms. However, the mechanism channels have not been explored. In this section, we investigate how supply chain network structure impacts systemic tail risk through channels such as information efficiency and risk diversification. Furthermore, we conduct heterogeneity analysis to explore nuanced patterns in the relationship between supply chain network structure and systemic tail risk.

5.1. Mechanism analysis

5.1.1. Information efficiency channel

Social networks help aggregate and disseminate various types of information, where firms with more robust supply chain network structure can access superior information resources within these networks (Chuluun et al., 2014), thereby enhancing the informational content of their stock prices. Consequently, these firms tend to be less vulnerable to market shocks during "black swan" events, leading to lower systemic tail risks. In this study, we use stock synchronicity (Syn) as a measure of firm information efficiency, following the methodology of Durnev et al. (2003). Specifically, we obtain R-squared from regressions of weekly individual stock returns on market

[6] In our study, if unobserved factors reduce systemic tail risk, we can randomly match the company's supply chain network structure indicators with other companies and repeat the test. Even then, we would still observe that an increase in supply network centrality suppresses systemic tail risk; if not, we can rule out the interference of other potential factors.

Table 11
Mechanism check.

	(1) *Syn*	(2) *Tr*	(3) *Dd*	(4) *Tr*
Sns	−0.151***	−0.425***	2.840***	−0.432***
	(−3.30)	(−4.30)	(3.80)	(−4.29)
Syn		0.157***		
		(6.47)		
Dd				−0.008***
				(−4.33)
Size	0.020***	−0.014	−0.364***	−0.013
	(3.01)	(−0.92)	(−3.08)	(−0.85)
Mb	0.000	−0.002	0.100***	−0.002
	(0.30)	(−0.88)	(7.04)	(−0.77)
Roe	−0.015	−0.035	−0.548***	−0.054
	(−1.10)	(−1.02)	(−3.12)	(−1.58)
Price	−0.021***	0.031*	1.170***	0.036**
	(−2.95)	(1.85)	(9.20)	(2.09)
Pastret	0.018***	−0.070***	−0.437***	−0.069***
	(4.71)	(−7.39)	(−7.84)	(−7.26)
Illiq	−0.254***	0.315***	4.385***	0.272**
	(−4.57)	(2.66)	(5.26)	(2.33)
Ivol	−0.822***	0.186***	−24.003***	−0.117*
	(−31.92)	(3.34)	(−58.80)	(−1.73)
Cons	0.326**	0.727**	23.790***	0.952***
	(2.27)	(2.21)	(9.62)	(2.88)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
R^2	0.636	0.604	0.821	0.602
adj. R^2	0.554	0.513	0.781	0.512
N	9338	9338	9352	9352
Sobel-Z value		−2.94		−2.86

This table reports the results of the mechanism analysis of the information efficiency and risk diversification. *Tr* is a systemic tail risk indicator for the month at the year-end. *Sns* is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

and industry returns to quantify stock synchronicity. Table 11 presents the empirical results. Columns (1) and (2) use supply chain network structure as explanatory variables and stock synchronicity and systemic tail risk as dependent variables, respectively. Column (1) shows a significant negative coefficient for *Sns*, suggesting that stronger supply chain network structure is associated with lower stock price synchronicity, which indicates higher information efficiency. In column (2), after including the stock synchronicity indicator, the coefficient for *Syn* is significantly positive, implying that higher stock synchronicity is associated with greater systemic tail risk. In addition, the *Sns* coefficient remains significantly negative at the 1 % level, demonstrating that enhanced supply chain network structure mitigates systemic tail risks by improving firm information efficiency. Furthermore, the Sobel Z has a value of −2.94 that is significant at the 1 % level, verifying partial mediation effects in the mechanism of information efficiency.

5.1.2. Risk diversification channel

Social networks help facilitate risk sharing among firms (Fafchamps and Gubert, 2007; Larcker et al., 2013). Firms with stronger supply chain network structure can access more resources, which is particularly crucial when firms face cash flow constraints or inadequate credit allocation. The support from network members allows firms to better disperse individual default risks. Moreover, firms with lower individual risks are less susceptible to large investor sell-offs during extreme market events, resulting in lower systemic tail risks. To measure firm default risk,[7] we use default distance (*Dd*), calculated following Bharath and Shumway's (2008) simplified algorithm of the Merton Distance to Default (DD) model. Table 11 presents the empirical results of the risk diversification test, with columns (3) and (4) using supply chain network structure as explanatory variables, and firm default distance and systemic tail risk as dependent variables. Column (3) shows a significantly positive coefficient for *Sns*, indicating that stronger supply chain network structure is associated with greater firm default distance, and hence lower default risk. When adding firm default distance into column (4), the coefficient for *Dd* is significantly negative, suggesting that greater default distance is associated with reduced systemic tail risk. Moreover, *Sns* exhibits a significantly negative coefficient at the 1 % significance level, indicating that higher supply chain network structure mitigates systemic tail risk by dispersing individual default risks. Additionally, the Sobel Z value indicates the presence of partial mediation effects in the mechanism of risk diversification.

[7] In large samples, the calculated P_{Merton} default probability does not align with the actual default probability, which is derived from empirical distributions, whereas P_{Merton} is obtained under the assumption of a normal distribution. Using default distance *Dd* can mitigate the calculation inconsistency arising from different distributions.

Table 12
Heterogeneity test: impact of capital market intermediaries.

	(1) Analyst	(2)	(3) Analyst report	(4)	(5) News	(6)
	Low	High	Low	High	Low	High
	Tr	*Tr*	*Tr*	*Tr*	*Tr*	*Tr*
Sns	−0.121	−0.499***	−0.103	−0.540***	−0.019	−0.415***
	(−0.48)	(−4.40)	(−0.45)	(−4.66)	(−0.09)	(−3.69)
Size	−0.027	−0.039	−0.017	−0.035	−0.021	0.008
	(−1.06)	(−1.63)	(−0.70)	(−1.44)	(−0.82)	(0.37)
Mb	−0.001	−0.003	−0.001	−0.001	−0.004	−0.007*
	(−0.37)	(−0.68)	(−0.37)	(−0.27)	(−1.05)	(−1.77)
Roe	−0.066	0.079	−0.061	0.107	−0.020	−0.106**
	(−1.59)	(1.15)	(−1.45)	(1.43)	(−0.34)	(−2.26)
Price	0.034	0.030	0.013	0.018	−0.002	0.030
	(1.06)	(1.39)	(0.44)	(0.82)	(−0.07)	(1.27)
Pastret	−0.100***	−0.043***	−0.095***	−0.046***	−0.104***	−0.041***
	(−5.25)	(−3.97)	(−5.33)	(−4.04)	(−6.31)	(−3.57)
Illiq	0.008	0.344*	0.036	0.331	−0.184	0.393**
	(0.05)	(1.66)	(0.23)	(1.53)	(−0.94)	(2.11)
Ivol	0.061	0.102	0.097	0.056	0.172**	−0.040
	(0.85)	(1.20)	(1.37)	(0.62)	(2.20)	(−0.51)
Cons	1.484***	1.769***	1.294***	1.721***	1.421***	0.821*
	(2.88)	(3.50)	(2.65)	(3.31)	(2.68)	(1.71)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.399	0.369	0.405	0.365	0.473	0.320
adj. R^2	0.396	0.366	0.402	0.362	0.471	0.317
N	4405	5009	4601	4813	4704	4710
Bdiff		0.377**		0.437**		0.396**

This table reports the results of heterogeneity analysis considering the impact of capital market intermediaries on information dissemination. The dependent variable is *Tr*, which is a systemic tail risk indicator for the month at the year-end. The independent variable is *Sns*, which is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

5.2. Heterogeneity analyses

5.2.1. Impact of capital market information intermediaries

Attention from analysts, news media, and other information intermediaries facilitates the dissemination of firm-specific information, which helps external investors understand corporate operational conditions in a timely manner. This study posits that heightened attention from information intermediaries enhances the role of supply chain network structure in improving corporate information efficiency, thereby exerting a stronger effect on reducing systemic tail risks. Specifically, we use analyst coverage (*Ana_num*, the natural logarithm of the number of analysts tracking the firm annually), analyst report coverage (*Ana_report*, the natural logarithm of the number of analyst reports published annually), and media coverage (*News*, the natural logarithm of the number of financial news articles annually sourced from CNRDS) to characterize attention from capital market information intermediaries. We divide the total sample into high and low groups based on the median value. Table 12 reports the findings and reveals that in groups with higher analyst coverage, analyst report coverage, and media coverage, the coefficient for *Sns* is significantly negative at the 1 % level. Conversely, in the low information intermediary attention groups, these coefficients are not significant. The significant differences between the high and low groups verify our conjecture.

5.2.2. Impact of corporate accounting information quality

Accounting information reflects a company's current financial and operating conditions, serving as a crucial source for investors to obtain firm-specific information. High-quality accounting information can convey more accurate firm-specific information to capital markets, thereby reducing stock price noise. This study posits that high-quality accounting information enhances the role of supply chain network structure in improving corporate information efficiency, thereby reinforcing its effect in reducing systemic tail risks. Specifically, we measure corporate accounting information quality based on earnings management (*Em*, manipulative accruals calculated using the adjusted Jones model) and annual report readability (*Readability*, textual readability of financial reports sourced from the WINGO database). The sample is divided into high and low groups based on their median values. Table 13 presents these findings, indicating that the coefficients for *Sns* are significantly negative at the 1 % or 10 % level in both groups. However, the coefficients are significantly lower in the low earnings management group and high annual report readability group (indicating higher accounting information quality), suggesting that in these groups supply chain network structure exhibits a larger mitigating effect on systemic tail risks.

Table 13
Heterogeneity test: impact of corporate accounting information quality.

	(1)	(2)	(3)	(4)
	Em		Readability	
	Low	High	Low	High
	Tr	*Tr*	*Tr*	*Tr*
Sns	−0.625***	−0.310*	−0.405***	−0.646***
	(−5.08)	(−1.84)	(−3.04)	(−3.98)
Size	−0.007	−0.012	−0.011	−0.010
	(−0.25)	(−0.57)	(−0.45)	(−0.39)
Mb	−0.009*	−0.001	0.000	−0.002
	(−1.76)	(−0.36)	(0.02)	(−0.67)
Roe	−0.092	−0.046	−0.152***	0.057
	(−1.22)	(−1.03)	(−3.08)	(1.16)
Price	0.062**	0.007	0.017	0.009
	(2.28)	(0.29)	(0.71)	(0.36)
Pastret	−0.044***	−0.077***	−0.056***	−0.072***
	(−2.98)	(−5.72)	(−4.26)	(−5.45)
Illiq	0.049	0.239	0.387**	−0.063
	(0.26)	(1.30)	(1.97)	(−0.42)
Ivol	0.096	−0.046	0.184**	−0.130
	(1.18)	(−0.58)	(2.43)	(−1.60)
Cons	1.013*	1.284***	1.115**	1.272**
	(1.82)	(2.83)	(2.14)	(2.52)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
R^2	0.384	0.401	0.364	0.409
adj. R^2	0.381	0.398	0.361	0.406
N	4608	4624	4684	4684
Bdiff		−0.315*		0.242*

This table reports the results of heterogeneity analysis considering the impact of corporate accounting information quality. The dependent variable is *Tr*, which is a systemic tail risk indicator for the month at the year-end. The independent variable is *Sns*, which is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

5.2.3. *Impact of customer or supplier geographic distance*

When firms are located closer to their customers or suppliers, financial transactions and resource flow within the supply chain tend to be more rapid. This study posits that shorter geographic distances between customers or suppliers enhance supply chain network structure in mitigating default risks, thereby reinforcing its effect in reducing systemic tail risks. Specifically, we use the median of the natural logarithm of mean distances between firms and their customers or suppliers to divide the sample into high and low groups. Table 14 presents these findings, indicating that *Sns* coefficients are only significantly negative in the low customer or supplier distance group, supporting our assumption.

5.2.4. *Impact of corporate financing constraints*

Companies with lower degree of financing constraints face fewer liquidity crises within the supply chain. This study posits that fewer financing constraints enhance supply chain network structure in mitigating debt default risks and thus systemic tail risks. Specifically, we use the FC index and WW index to measure the extent of corporate financing constraints. The sample is divided into high and low groups based on the median values of these variables, where smaller FC and WW indices indicate lower financing constraints. Table 14 reports the findings, indicating that *Sns* coefficients are significantly negative at the 1 % level in the low financing constraints group, while they are not significant in the high financing constraints group. Moreover, there are significant differences of the coefficient between the high and low groups, verifying our conjecture.

5.2.5. *Impact of supply chain transparency and extreme market conditions*

Supply chain transparency reflects the degree to which external stakeholders understand a company's supply chain network structure, which may influence our conclusions. Therefore, we conduct a heterogeneity analysis based on supply chain-related characteristics. Specifically, we measure enterprise supply chain transparency using the number of large suppliers and customers with disclosed names (*Sct_Num*) and the proportion of transactions from large suppliers and customers with disclosed names to the total transactions of the top five suppliers and customers (*Sct_Ratio*). Based on the median of the sample, we divide the full sample into high and low transparency groups. Table 15 shows that when a company's supply chain transparency is higher, the centrality of the supply network has a more significant effect on mitigating systemic tail risk.

Furthermore, considering the impact of supply network centrality on systemic tail risk may vary under extreme market conditions, such as bear markets, we perform a heterogeneity analysis based on market environment-related characteristics. Table 15 shows that when companies are in extreme market conditions (i.e., during the 2015 financial crisis or the more severe periods of the COVID-19 pandemic in 2020), the centrality of the supply network has a more pronounced effect on mitigating systemic tail risk. This reflects the

Table 14
Heterogeneity test: impact of customer/supplier geographic distance or firm financing constraints.

	(1) Distance	(2)	(3) Fc	(4)	(5) Ww	(6)
	Low *Tr*	High *Tr*	Low *Tr*	High *Tr*	Low *Tr*	High *Tr*
Sns	−0.602***	−0.206	−0.411***	0.125	−0.421***	0.110
	(−3.90)	(−1.14)	(−3.81)	(0.29)	(−3.91)	(0.29)
Size	0.041	−0.062*	−0.007	−0.035	−0.027	−0.035
	(1.28)	(−1.95)	(−0.27)	(−1.34)	(−1.08)	(−1.32)
Mb	−0.008*	0.000	−0.009**	0.006	−0.009*	0.001
	(−1.72)	(0.12)	(−2.34)	(1.59)	(−1.74)	(0.23)
Roe	−0.021	−0.075	−0.093**	0.041	0.059	−0.056
	(−0.39)	(−1.22)	(−2.20)	(0.68)	(0.78)	(−1.30)
Price	0.035	0.051	0.037	0.018	0.056**	−0.019
	(1.11)	(1.48)	(1.37)	(0.77)	(2.21)	(−0.72)
Pastret	−0.016	−0.059***	−0.076***	−0.058***	−0.047***	−0.101***
	(−1.00)	(−2.83)	(−5.20)	(−3.99)	(−3.52)	(−6.59)
Illiq	0.270	0.323	−0.009	0.231	0.075	0.159
	(1.32)	(1.48)	(−0.04)	(1.56)	(0.31)	(1.09)
Ivol	0.042	0.013	0.086	0.058	0.052	0.106
	(0.46)	(0.14)	(1.14)	(0.75)	(0.68)	(1.23)
Cons	0.046	2.305***	1.077**	1.642***	1.491***	1.753***
	(0.07)	(3.53)	(2.02)	(3.05)	(2.73)	(3.25)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.382	0.356	0.361	0.424	0.374	0.428
adj. R^2	0.378	0.351	0.358	0.422	0.371	0.425
N	2928	2929	4621	4622	4420	4420
Bdiff		−0.396*		−0.536***		−0.531***

This table reports the results of heterogeneity analysis considering both the impact of customer/supplier geographic distance and firm financing constraints. The dependent variable is *Tr*, which is a systemic tail risk indicator for the month at the year-end. The independent variable is *Sns*, which is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

fact that under extreme market pressures, companies with highly centralized supply networks, through their resource integration capabilities and status as information hubs, are able to more effectively coordinate the flow of resources within the industry chain, thereby mitigating risk contagion.

6. Conclusion

This study investigates how supply chain network structure influences financial stability in the capital market. We construct the supply chain network structure using data from the annual reports of Chinese A-share listed companies from 2009 to 2022, focusing on the top five customers and suppliers. By measuring the network structure, we examine its impact on financial stability. Our empirical results suggest that a stronger supply chain network structure is associated with lower systemic tail risk in the future. We further demonstrate that this effect operates through mechanisms of enhanced information efficiency and risk diversification. Additionally, the effect is more pronounced under conditions of greater attention from information intermediaries, higher quality accounting disclosures, closer proximity to customers or suppliers, and lower financing constraints.

Our findings have practical implications for both governments and firms. This research provides micro-level evidence of the economic effects of supply chain network structure and suggests that enhancing supply chain network structure contributes to financial market stability. To build a robust supply chain network structure, managers should foster synergies among firms within the network through contractual arrangements that promote collaboration. Meanwhile, policymakers should focus on facilitating the efficient flow of information and resources within the network.

Author statement

All the authors declare that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. There is no conflict of interest within our paper submission. And the manuscript is approved by all authors for publication.

CRediT authorship contribution statement

Rongsheng Zhuo: Visualization, Software, Methodology, Data curation. **Ao Li:** Writing – review & editing, Writing – original draft,

Table 15
Heterogeneity test: impact of supply chain transparency or market conditions.

	(1)	(2)	(3)	(4)	(5)	(6)
	Sct_num		Sct_ratio		Market_condition	
	Low	High	Low	High	Good	Bad
	Tr	*Tr*	*Tr*	*Tr*	*Tr*	*Tr*
Sns	0.093	−0.634***	−0.036	−0.592***	−0.459***	−1.130***
	(0.36)	(−5.55)	(−0.18)	(−5.28)	(−4.33)	(−3.10)
Size	−0.071	−0.017	−0.093**	−0.011	−0.006	−0.081
	(−1.56)	(−0.95)	(−2.19)	(−0.63)	(−0.38)	(−1.61)
Mb	−0.001	−0.004	0.000	−0.004	−0.005	0.010**
	(−0.14)	(−1.56)	(0.06)	(−1.35)	(−1.59)	(2.25)
Roe	−0.006	−0.049	−0.007	−0.053	−0.043	−0.045
	(−0.08)	(−1.23)	(−0.09)	(−1.35)	(−1.13)	(−0.46)
Price	0.049	0.028	0.053	0.029	0.018	0.155***
	(1.24)	(1.37)	(1.40)	(1.44)	(1.01)	(3.17)
Pastret	−0.043**	−0.073***	−0.042**	−0.073***	−0.081***	0.030
	(−2.31)	(−6.48)	(−2.33)	(−6.50)	(−7.32)	(0.82)
Illiq	0.722	0.149	0.313	0.104	0.322**	0.560*
	(1.59)	(1.19)	(0.91)	(0.82)	(2.14)	(1.77)
Ivol	0.229**	0.032	0.246**	0.014	0.145**	−0.606***
	(2.00)	(0.54)	(2.43)	(0.24)	(2.46)	(−3.33)
Cons	2.383**	1.311***	2.880***	1.189***	1.044***	2.167**
	(2.46)	(3.55)	(3.23)	(3.25)	(2.99)	(2.05)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.335	0.376	0.344	0.382	0.383	0.397
adj. R^2	0.329	0.374	0.338	0.380	0.382	0.392
N	2063	7351	2280	7134	8210	1204
Bdiff		0.727***		0.556**		0.671*

This table reports the results of heterogeneity analysis considering both the impact of supply chain transparency or market conditions. The dependent variable is *Tr*, which is a systemic tail risk indicator for the month at the year-end. The independent variable is *Sns*, which is supply network structure, measured by the natural logarithm of the weighted degree centrality of nodes in a supply network. Other variable definitions are in Table 2. The standard errors are clustered at the firm level. *, ** and *** represent statistical significance at the 10 %, 5 % and 1 % levels, respectively.

Validation, Methodology. **Zheng Qiao:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Yu Cai:** Visualization, Project administration, Methodology. **Bo Sang:** Writing – review & editing, Visualization, Supervision, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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